MRPL — INCIDENT INVESTIGATION REPORT

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Incident Ref:	INC-2024-0312-P101
Incident Date:	12-March-2024, 14:35 hrs
Location:	Process Area 3, Pump Bay — P-101
Incident Type:	Equipment Failure — Bearing Failure with Secondary Fire Risk
Severity:	High — Production Impact + Potential Safety Risk
Investigated By:	Safety Officer — Priya M., Lead Investigator
Report Date:	25-March-2024
Distribution:	Maintenance Manager, Plant Manager, Safety Committee

1. INCIDENT DESCRIPTION

At approximately 14:35 hrs on 12-March-2024, Pump P-101 (Crude Oil Feed Pump, Process Area 3) suffered a catastrophic drive-end (DE) bearing failure. The bearing seizure caused the shaft to deflect, resulting in mechanical seal failure and subsequent crude oil release. The released crude oil contacted the hot bearing housing (surface temperature approximately 220°C at failure) and ignited, causing a small fire in the pump bay.

The fire was contained within 3 minutes by the plant fire team using CO_2 extinguishers. No personnel injuries were sustained. Estimated production loss: 8 hours (2,560 m^3 crude throughput). Equipment damage: Pump P-101 required complete rebuild.

2. TIMELINE OF EVENTS

Time	Event
06:00, 12-Mar	Shift handover — P-101 reported normal. DE bearing temp 82°C (DCS).
09:15	DE bearing temp alarm activated at 85°C. Control room acknowledged.
09:20	Shift supervisor notified. Decision to monitor (no action taken per operator judgment).
11:30	Second bearing temp alarm — 88°C. Technician dispatched.
11:45	Technician inspection: pump noisy but continuing to run. Decision to keep running until next shift.
13:00	DE bearing temp reaches 95°C — DCS trip signal generated. Trip was bypassed by control room (INCORRECT AC
14:35	**Bearing seizure. Shaft deflection. Seal failure. Crude oil release. Fire ignition.**
14:38	Fire alarm activated. Fire team response. ESD initiated.
14:41	Fire extinguished. Area evacuated and secured.

3. ROOT CAUSE ANALYSIS

3.1 Immediate Cause

Catastrophic failure of DE bearing SKF 6316 due to lubrication starvation and metal fatigue. Post-failure analysis of bearing debris confirmed: (a) Complete grease oxidation — grease had turned to dry carbon coke, providing no lubrication. (b) Severe spalling on outer race over approximately 60% of the bearing raceway. (c) Roller cage fracture due to metal fatigue.

3.2 Contributing Causes

(a) MISSED REGREASING: SAP PM records confirm that the last regreasing of P-101 DE bearing was performed 18 months before the incident — 12 months beyond the required 6-month interval. Work Order WO-2022-P101-REG was closed without execution due to plant outage scheduling conflict and was not rescheduled.

(b) ALARM MANAGEMENT FAILURE: The 85°C bearing temperature alarm was acknowledged but not acted upon from 09:15 to 13:00 (3 hours 45 minutes). This is a violation of SOP-PM-007 Section 4.3 which requires controlled shutdown if bearing temperature is above 85°C and rising.

(c) DCS TRIP BYPASS: At 13:00, the DCS high-high temperature trip (95°C) was bypassed by the control room operator without authorisation. This is a critical safety violation. The bypass removed the last automatic protective layer that would have prevented the bearing seizure.

3.3 Root Cause

Root cause: Inadequate Maintenance Management System. Specifically, the absence of a robust system to ensure overdue preventive maintenance is rescheduled and escalated when not completed on time. The missed regreasing was the primary initiating event.

4. CORRECTIVE AND PREVENTIVE ACTIONS (CAPA)

#	Action	Responsible	Due Date	Status
1	Bearing regreasing overdue >30 days: mandatory escalation to Maintenance Manager.	Maint. Mgr.	01-Apr-2024	CLOSED
2	DCS protection bypass: require dual authorisation (Shift Supervisor + Safety).	Instrument Engr.	15-Apr-2024	CLOSED
3	Retrain all operators on SOP-PM-007 Section 4.3 alarm response.	HSE Dept.	30-Apr-2024	CLOSED
4	Install continuous online vibration monitor on P-101 DE bearing.	Project Engr.	30-Jun-2024	IN PROGRESS
5	Reduce bearing temperature alarm from 85°C to 80°C for P-101.	Instrument Engr.	01-Apr-2024	CLOSED
6	Monthly audit of overdue PM work orders for critical rotating equipment.	Maint. Mgr.	Ongoing	ACTIVE

5. LESSONS LEARNED

1. Never ignore a bearing temperature alarm on crude oil service pumps. The alarm-to-failure window can be as short as 5 hours when lubrication is severely degraded.

2. DCS protective trips must never be bypassed without full safety review and dual authorisation. Protective systems are the last line of defence against catastrophic failure.

3. Overdue preventive maintenance on critical equipment must trigger automatic escalation. Manual tracking is insufficient for high-consequence items.

4. Lubrication starvation is the leading cause of bearing failure on centrifugal pumps in hydrocarbon service. Strict adherence to regreasing intervals is essential.

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